



MHH772 / 90DaysOfDevOps



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90DaysOfDevOps / 2026 / day-11 /



LondheShubham153 Added day 11 tasks

68e1bfe · 7 months ago



This branch is 10 commits ahead of TrainWithShubham/90DaysOfDevOps:master .

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Sync fork

Name	Name	Last commit date
..		
README.md	Added day 11 tasks	7 months ago

README.md



Day 11 – File Ownership Challenge (chown & chgrp)

Task

Master file and directory ownership in Linux.

- Understand file ownership (user and group)
- Change file owner using `chown`
- Change file group using `chgrp`
- Apply ownership changes recursively

Expected Output

- A markdown file: `day-11-file-ownership.md`
- Screenshots showing ownership changes

Challenge Tasks

Task 1: Understanding Ownership (10 minutes)

1. Run `ls -l` in your home directory
2. Identify the **owner** and **group** columns
3. Check who owns your files

Format: `-rw-r--r-- 1 owner group size date filename`

Document: What's the difference between owner and group?

Task 2: Basic chown Operations (20 minutes)

1. Create file `devops-file.txt`
2. Check current owner: `ls -l devops-file.txt`
3. Change owner to `tokyo` (create user if needed)
4. Change owner to `berlin`
5. Verify the changes

Try:

```
sudo chown tokyo devops-file.txt
```



Task 3: Basic chgrp Operations (15 minutes)

1. Create file `team-notes.txt`
2. Check current group: `ls -l team-notes.txt`
3. Create group: `sudo groupadd heist-team`

4. Change file group to `heist-team`
5. Verify the change

Task 4: Combined Owner & Group Change (15 minutes)

Using `chown` you can change both owner and group together:

1. Create file `project-config.yaml`
2. Change owner to `professor` AND group to `heist-team` (one command)
3. Create directory `app-logs/`
4. Change its owner to `berlin` and group to `heist-team`

Syntax: `sudo chown owner:group filename`

Task 5: Recursive Ownership (20 minutes)

1. Create directory structure:

```
mkdir -p heist-project/vault
mkdir -p heist-project/plans
touch heist-project/vault/gold.txt
touch heist-project/plans/strategy.conf
```



2. Create group `planners` : `sudo groupadd planners`
3. Change ownership of entire `heist-project/` directory:
 - Owner: `professor`
 - Group: `planners`
 - Use recursive flag (`-R`)
4. Verify all files and subdirectories changed: `ls -lR heist-project/`

Task 6: Practice Challenge (20 minutes)

1. Create users: `tokyo` , `berlin` , `nairobi` (if not already created)
2. Create groups: `vault-team` , `tech-team`

3. Create directory: bank-heist/

4. Create 3 files inside:

```
touch bank-heist/access-codes.txt
touch bank-heist/blueprints.pdf
touch bank-heist/escape-plan.txt
```



5. Set different ownership:

- access-codes.txt → owner: tokyo , group: vault-team
- blueprints.pdf → owner: berlin , group: tech-team
- escape-plan.txt → owner: nairobi , group: vault-team

Verify: `ls -l bank-heist/`

Key Commands Reference

```
# View ownership
ls -l filename
```



```
# Change owner only
sudo chown newowner filename
```

```
# Change group only
sudo chgrp newgroup filename
```

```
# Change both owner and group
sudo chown owner:group filename
```

```
# Recursive change (directories)
sudo chown -R owner:group directory/
```

```
# Change only group with chown
sudo chown :groupname filename
```

Hints

- Most `chown` / `chgrp` operations need `sudo`
- Use `-R` flag for recursive directory changes

- Always verify with `ls -l` after changes
- User must exist before using in `chown`
- Group must exist before using in `chgrp` / `chown`

Documentation

Create `day-11-file-ownership.md` :

Day 11 Challenge

Files & Directories Created

[list all files/directories]

Ownership Changes

[before/after for each file]

Example:

- `devops-file.txt`: user:`user` → tokyo:`heist-team`

Commands Used

[your commands here]

What I Learned

[3 key points about file ownership]



Troubleshooting

Permission denied?

- Use `sudo` for `chown`/`chgrp` operations

Group doesn't exist?

- Create it first: `sudo groupadd groupname`

User doesn't exist?

- Create it first: `sudo useradd username`

Why This Matters for DevOps

In real DevOps scenarios, you need proper file ownership for:

- Application deployments
- Shared team directories
- Container file permissions
- CI/CD pipeline artifacts
- Log file management

Submission

1. Navigate to `2026/day-11/` folder
2. Add `day-11-file-ownership.md` with screenshots
3. Commit and push to your fork

Learn in Public

Share on LinkedIn about mastering file ownership.

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Happy Learning **TrainWithShubham**